INCH-POUND

MIL-C-24643/56D
22 November 1994
SUPERSEDING
MIL-C-24643/56C
14 March 1994

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS2SWL-7
(INCLUDING VARIATION LS2SWLA-7)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LS2SWL-7 (unarmored) and LS2SWLA-7 (armored).

REQUIREMENTS:

Qualification required.Construction (watertight)

* First - Copper conductor, tin coated, AWG size 16, class B stranding.
Second - Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation. Nominal wall thickness 0.019 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black and one white conductor for each pair.
* Fourth - Two conductors (cabled together with a nominal lay of 3 inches to form a pair.
Fifth - Binder tape over each pair, applied at manufacturer's option, applied helically with overlap.
* Sixth - Braided shield of AWG No. 34 or No. 36 tin coated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:18Z
Check the source to verify that this is the current version before use.

Construction (watertight) (Continued)

Seventh- Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, outer tape sealed. An alternate shield insulation may consist of one tape, type G of MIL-I-631, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

Eighth - Seven shielded pairs cabled together with a lay not greater than 18 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers shall be used as necessary to make a firm, well-rounded assembly.

Ninth - Binder tape, applied helically with overlap.

Construction variant for type LS2SWL-7

Tenth - Cross-linked polyolefin jacket. Minimum average jacket thickness 0.075 inch. Cable overall diameter shall be 0.860 minimum, 0.910 maximum inch.

Construction variant for type LS2SWLA-7

Tenth - Cross-linked polyolefin jacket. Minimum average jacket thickness 0.075 inch. Cable overall diameter shall be 0.910 minimum, 0.960 maximum inch.

Eleventh - Braided metal armor.

* NOTES:

1. Variant LS2SWL-7 military part number - M24643/56-01UN.
2. Variant LS2SWLA-7 military part number - M24643/56-02AN.

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	4.66
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield (pair shields only)	500
Insulation resistance - Megohms/1000 feet minimum	
Conductor to shield	500
Component shield to shield	100
* Conductor and shield continuity	No failure

Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-C-24643 for water leakage	No failure

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:18Z
Check the source to verify that this is the current version before use.

EXAMINATION AND TESTS (Continued)	Requirements
Capacitance	
Mutual capacitance - 1 kilohertz (KHz), pF/ft, maximum	35
Capacitance unbalance - percent maximum	10
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200 °C)	50
Drip - 95 ±1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals - (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (jacket) - cycles minimum	125
Cable sealant removability	No failure
Armor - (Type LS2SWLA-7 only) - conformance to material, construction and coverage	No failure
Shield conformance to material, construction and coverage	No failure

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:18Z
Check the source to verify that this is the current version before use.

EXAMINATION AND TESTS (Continued)

	Requirements
Group D:	
Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ±5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: 1500 feet nominal

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MI
Navy - SH

Review Activities:
Army - AV, CR, ER, ME, AR, AL
Navy - EC, CG
DLA - IS

Preparing Activity:
Navy - SH

Agent:
DLA-IS

(Project No. 6145-2101-56)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:18Z
Check the source to verify that this is the current version before use.